

End-to-End Retail Performance and Behavioral Analytics

Abstract:

This project, titled “**End-to-End Retail Performance and Behavioral Analytics**,” This project analyzes the retail performance and customer behavior of Retail Nova Inc., a mid-sized retail and e-commerce company. The objective is to identify sales trends, customer insights, product performance, store profitability, and return patterns using data analytics.

The project includes data cleaning and exploratory analysis in Python, business metric calculations using SQL, and interactive dashboard creation in Power BI or Tableau. The findings support data-driven decision-making to improve profitability, optimize inventory, reduce returns, and enhance overall business performance.

INTRODUCTION

Retail organizations generate large volumes of transactional and customer data on a daily basis. However, without proper analytics, this data cannot be effectively utilized for decision making. Retail analytics enables businesses to identify sales trends, understand customer purchasing behavior, measure product performance, and optimize store operations.

In a competitive retail environment, organizations must rely on data-driven insights rather than intuition. This project simulates a real-world retail analytics scenario where sales, customer, product, store, and return data are analyzed to support strategic business decisions.

Project Objectives

The key objectives of this project are:

- Analyze total revenue and quantity sold over time
- Identify top-selling and most profitable products
- Understand customer distribution and demographics
- Evaluate store performance and profitability • Analyze return patterns and trends

Tools and Technologies:

Used Python (Pandas, Jupyter Notebook) was used for data cleaning and pre-processing, SQL Server was used for database management and analysis, and Power BI was used for data modelling and visualization.

Methodology:

1. Business understanding
2. Data collection and understanding
3. Data cleaning and transformation
4. Python-based calculations
5. SQL-based analysis
6. Power BI dashboard creation
7. Insight generation

Data collection and understanding

DATASET DESCRIPTION

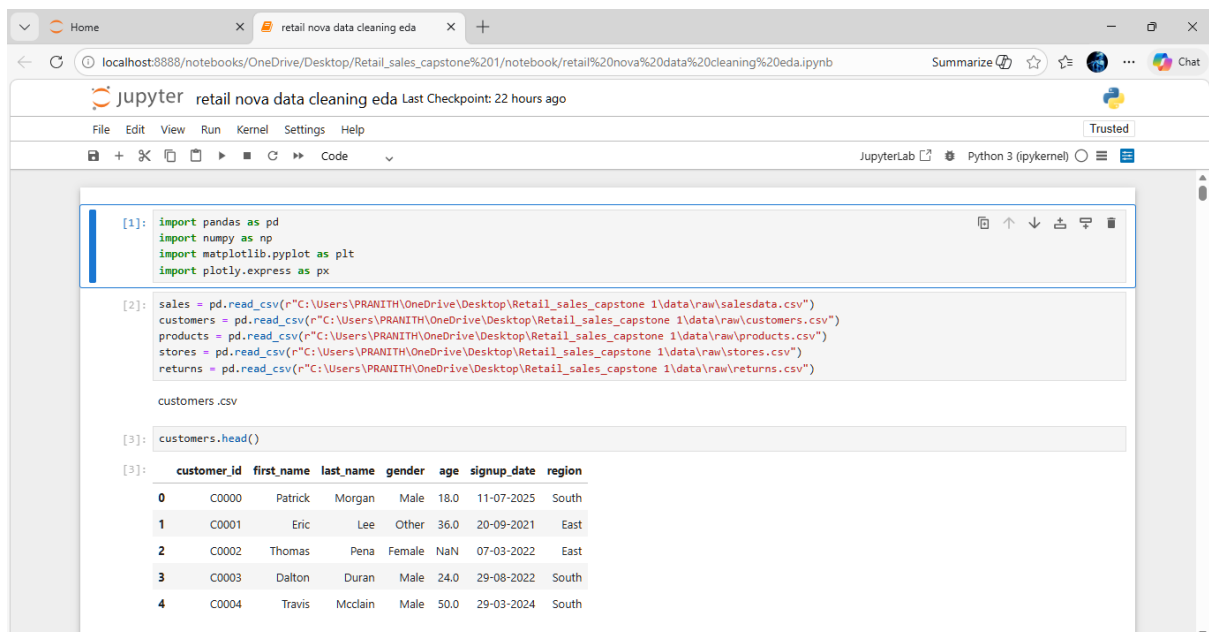
The project uses multiple datasets representing different business entities:

- **Sales Data:** Transaction details such as order date, quantity, unit price, discount, and total amount
- **Customer Data:** Customer demographics including age, gender, region, and signup date
- **Product Data:** Product attributes such as category, brand, cost price, and margin
- **Store Data:** Store details including location, type, and operating cost
- **Returns Data:** Returned orders with return date and reason. These datasets together provide a complete picture of retail operations.

DATA CLEANING & PREPARATION

Data Cleaning Process

Data cleaning was performed using SQL and Python. Unnecessary columns were removed, missing values were handled appropriately, and data types were standardized. Foreign key constraints were applied to maintain referential integrity across tables.



The screenshot shows a JupyterLab notebook titled "retail nova data cleaning eda". The code in the notebook performs the following steps:

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import plotly.express as px

[2]: sales = pd.read_csv(r"C:\Users\PRANITH\OneDrive\Desktop\Retail_sales_capstone 1\data\raw\salesdata.csv")
customers = pd.read_csv(r"C:\Users\PRANITH\OneDrive\Desktop\Retail_sales_capstone 1\data\raw\customers.csv")
products = pd.read_csv(r"C:\Users\PRANITH\OneDrive\Desktop\Retail_sales_capstone 1\data\raw\products.csv")
stores = pd.read_csv(r"C:\Users\PRANITH\OneDrive\Desktop\Retail_sales_capstone 1\data\raw\stores.csv")
returns = pd.read_csv(r"C:\Users\PRANITH\OneDrive\Desktop\Retail_sales_capstone 1\data\raw\returns.csv")

customers.csv

[3]: customers.head()
```

The output of the `customers.head()` command is displayed as a table:

	customer_id	first_name	last_name	gender	age	signup_date	region
0	C0000	Patrick	Morgan	Male	18.0	11-07-2025	South
1	C0001	Eric	Lee	Other	36.0	20-09-2021	East
2	C0002	Thomas	Pena	Female	NaN	07-03-2022	East
3	C0003	Dalton	Duran	Male	24.0	29-08-2022	South
4	C0004	Travis	Mcclain	Male	50.0	29-03-2024	South

Python-based calculations

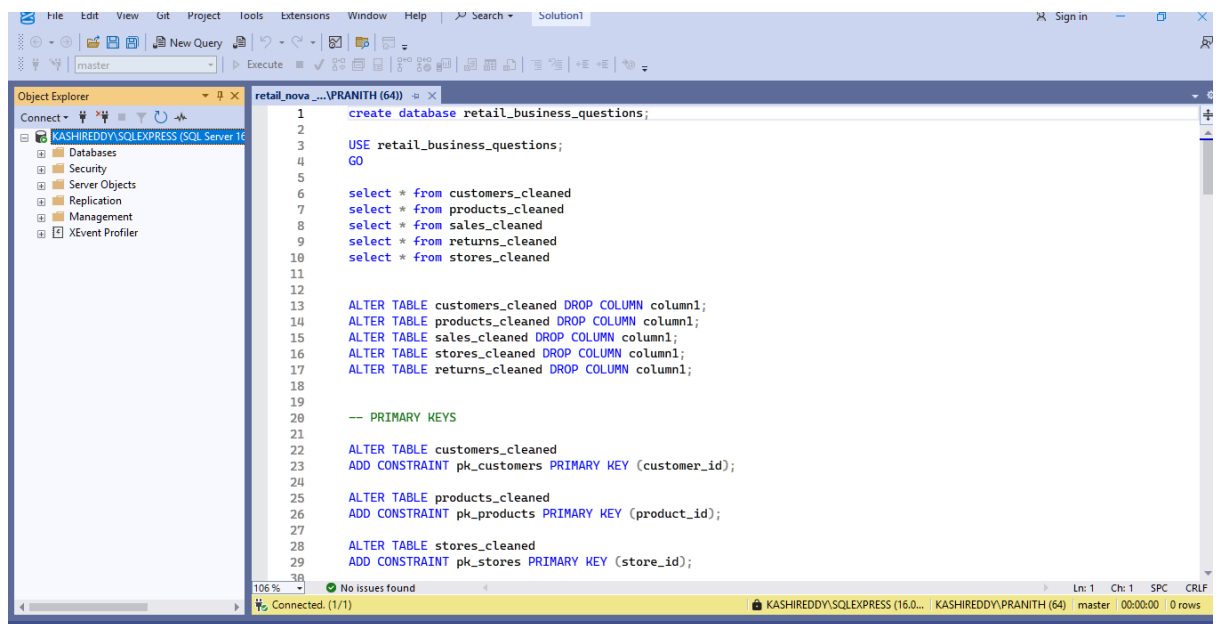
Profit Calculation

Profit was calculated at the transaction level using Python by subtracting total product cost from total sales amount. This profit value was stored in the sales table to ensure consistency across SQL analysis and Power BI dashboards.

SQL Analysis

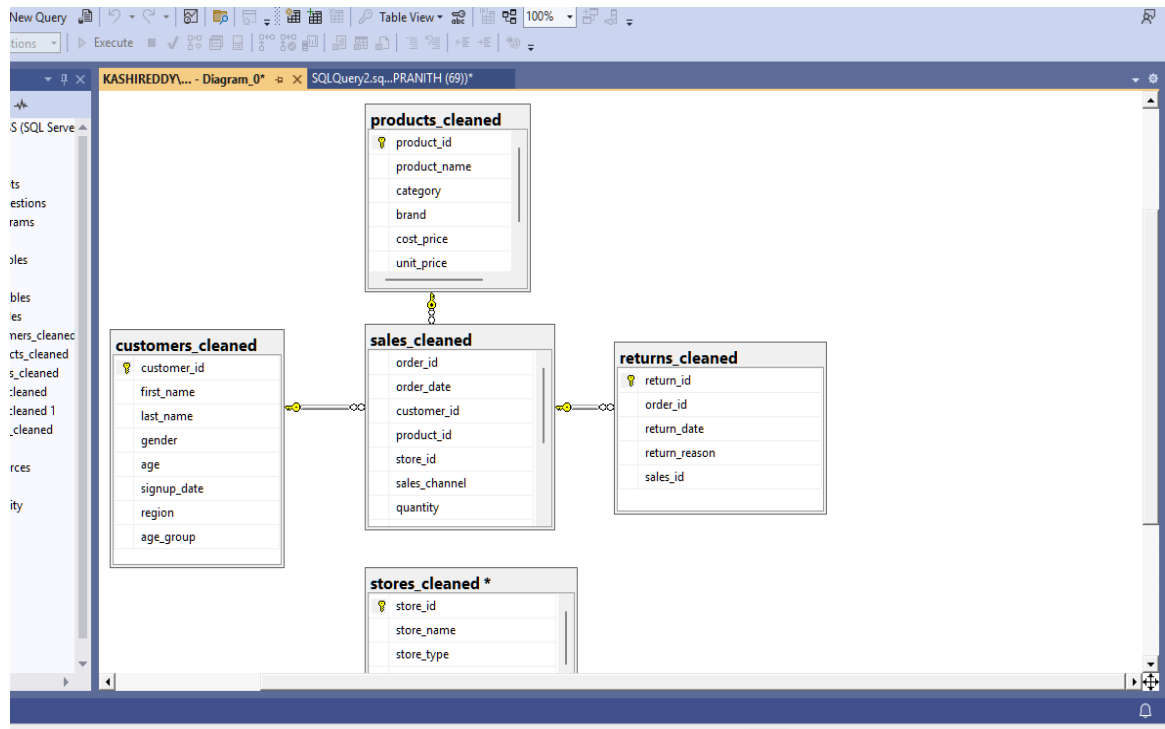
Cleaned data was loaded into Microsoft SQL Server for structured analysis. SQL queries were written to answer all mandatory business questions, including revenue trends, top performing products, customer distribution, return rates, and store profitability. Common SQL techniques used included joins, aggregations, window functions, and common table expressions (CTEs) to ensure clarity and performance.

DATA MODELING



```
1 create database retail_business_questions;
2
3 USE retail_business_questions;
4 GO
5
6 select * from customers_cleaned
7 select * from products_cleaned
8 select * from sales_cleaned
9 select * from returns_cleaned
10 select * from stores_cleaned
11
12
13 ALTER TABLE customers_cleaned DROP COLUMN column1;
14 ALTER TABLE products_cleaned DROP COLUMN column1;
15 ALTER TABLE sales_cleaned DROP COLUMN column1;
16 ALTER TABLE stores_cleaned DROP COLUMN column1;
17 ALTER TABLE returns_cleaned DROP COLUMN column1;
18
19
20 -- PRIMARY KEYS
21
22 ALTER TABLE customers_cleaned
23 ADD CONSTRAINT pk_customers PRIMARY KEY (customer_id);
24
25 ALTER TABLE products_cleaned
26 ADD CONSTRAINT pk_products PRIMARY KEY (product_id);
27
28 ALTER TABLE stores_cleaned
29 ADD CONSTRAINT pk_stores PRIMARY KEY (store_id);
30
```

ER_DIAGRAM

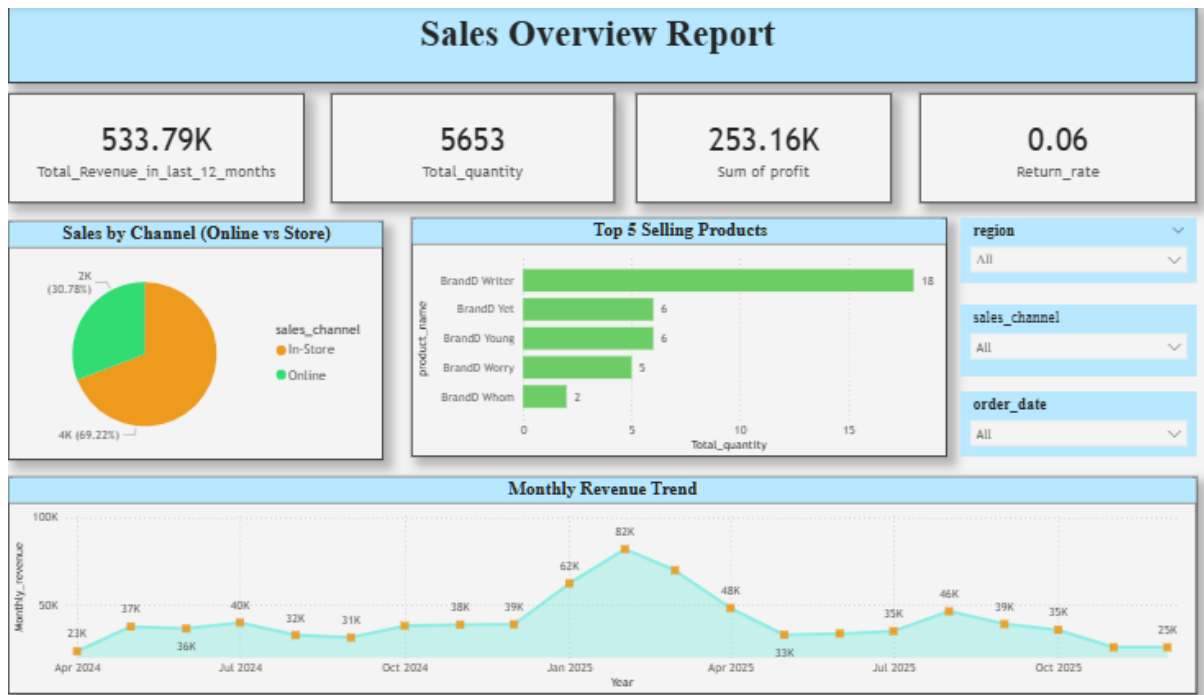


Explanation :-

A star-schema-based relational data model was created with the sales table acting as the central fact table. Dimension tables such as customers, products, stores, and returns are connected using primary and foreign key relationships. This structure ensures efficient querying, accurate aggregations, and seamless dashboard filtering.

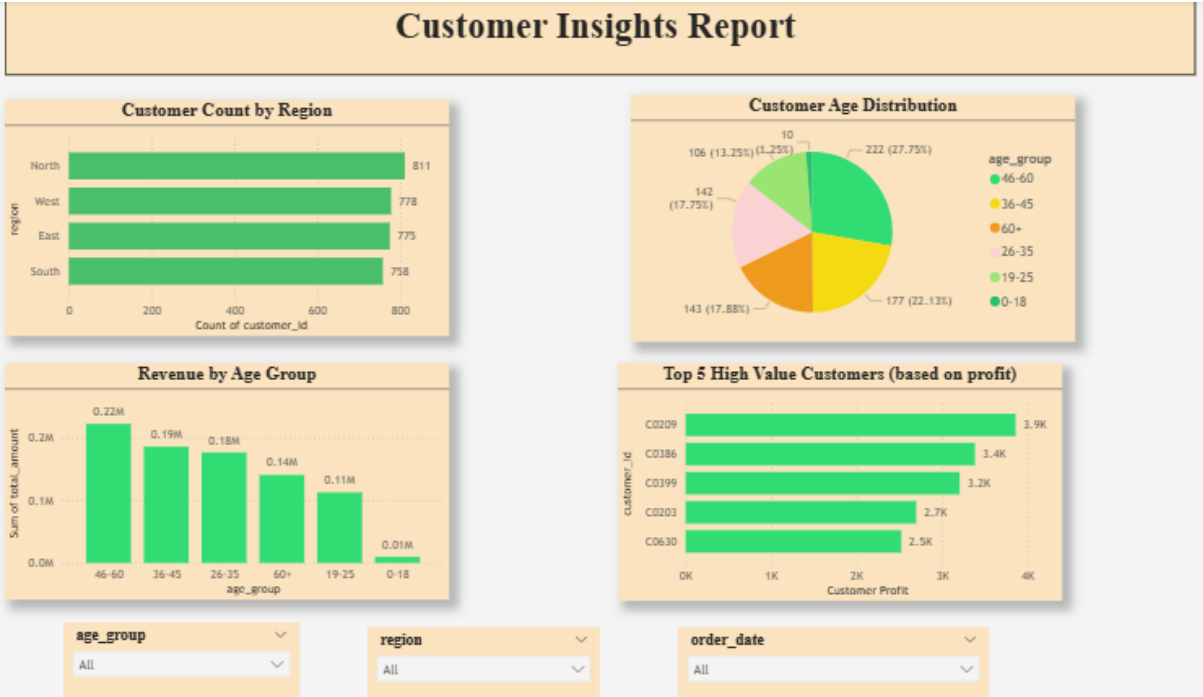
Power BI dashboard

SALES OVERVIEW DASHBOARD



The Sales Overview dashboard provides a high-level summary of business performance. It includes total revenue for the last 12 months, total quantity sold, monthly revenue trends, top 5 selling products, and sales by channel (Online vs In-Store). This dashboard helps stakeholders quickly assess overall sales health.

CUSTOMER INSIGHTS DASHBOARD



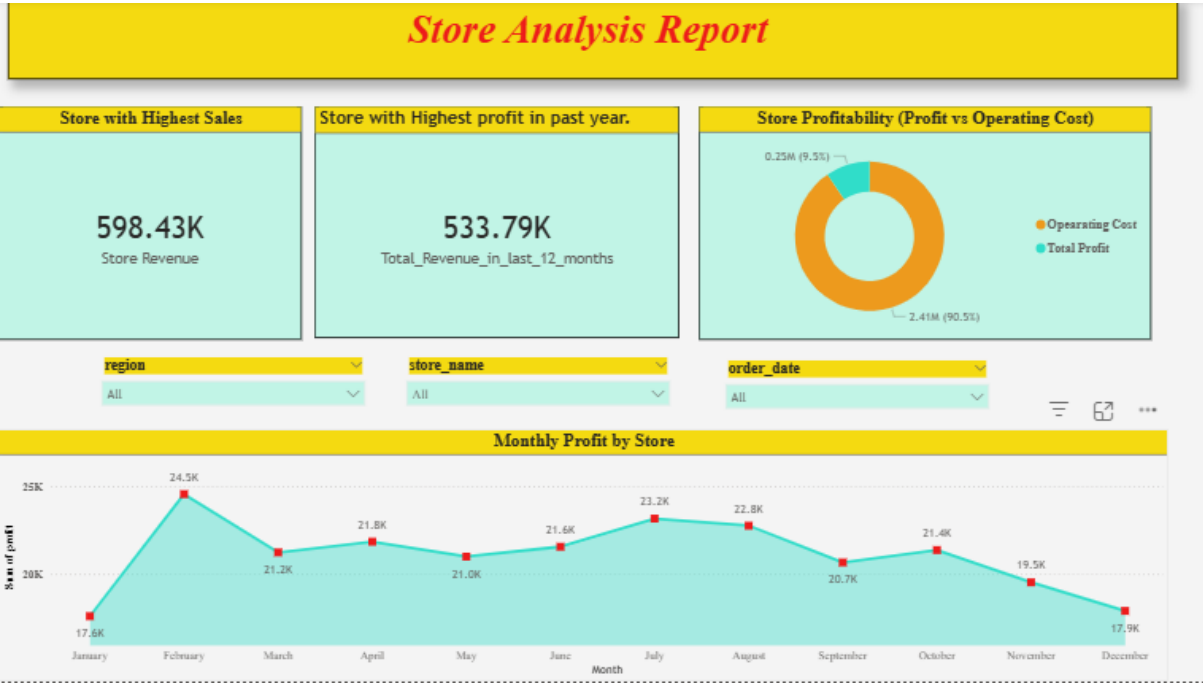
This dashboard focuses on customer behavior and demographics. It displays customer count by region, age distribution, revenue contribution by age group, and the top 5 high-value customers based on profit. These insights help businesses understand customer segments and target high-value customers effectively.

PRODUCT PERFORMANCE DASHBOARD



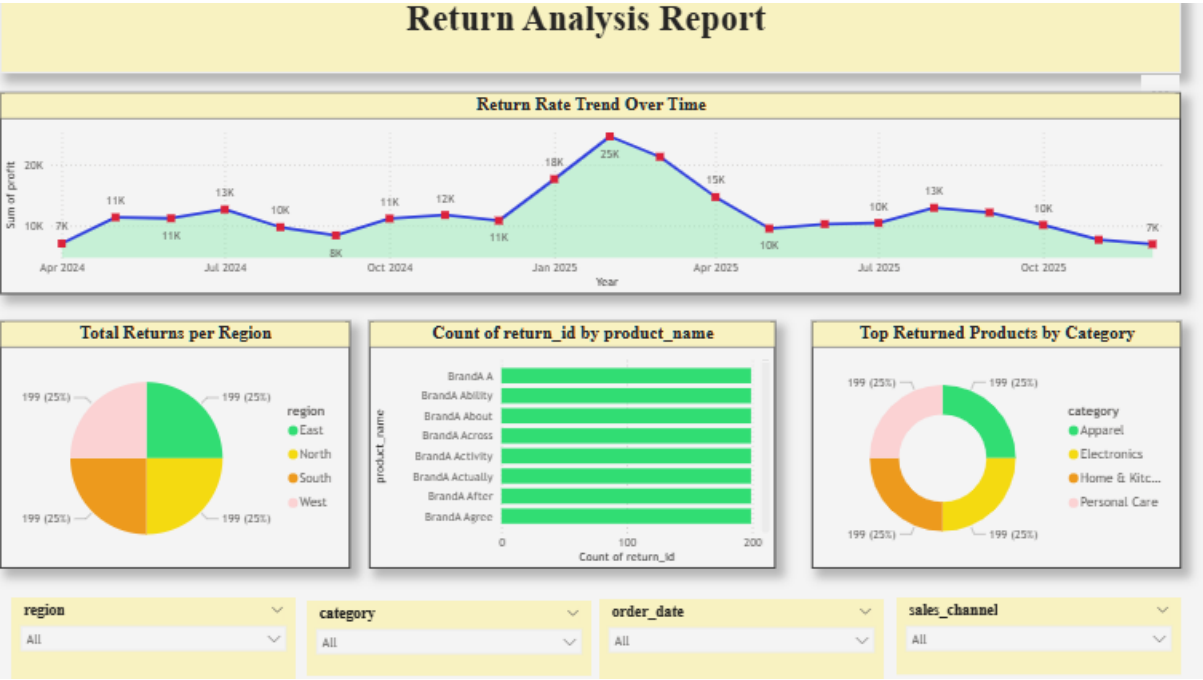
The Product Performance dashboard analyzes product-level metrics such as total profit by category, return rate by category, and top products by sales quantity. This dashboard helps identify profitable categories and products with high return rates that may require quality or supply chain improvements.

STORE ANALYSIS DASHBOARD



This dashboard evaluates store performance. It includes monthly profit by store, store profitability compared to operating cost, These insights help optimize store operations

RETURN ANALYSIS DASHBOARD



This dashboard evaluates return trends. It includes return rate trends over time, top returned products, and total returns by region. These insights help optimize store operations and reduce product returns.

Business Recommendations

- Focus marketing on high-value customers identified
- Reduce return rates in high-return categories
- Optimize store operating costs
- Improve quality checks for frequently returned products

CONCLUSION

This Retail Sales Analytics project demonstrates a complete end-to-end analytics workflow using SQL, Python, and Power BI. By integrating data cleaning, analysis, and visualization, the project delivers actionable insights that support data-driven decision-making. The dashboards provide a clear, interactive, and professional view of retail performance, making this project suitable for real-world business applications.